Telescopes: Types and How They Work

What you will be able to do

1. State the **three jobs** of a telescope, in order of importance.
2. Describe **refractors, reflectors and catadioptric** telescopes with their pros and cons.
3. Identify the parts of a beginner telescope: objective, eyepiece, focuser, finder, diagonal, Barlow, mount.
4. Compare **alt-azimuth, Dobsonian, equatorial and GoTo** mounts.
5. Know India's major research telescopes.

1. A short history

Year	Event
1608	Hans Lipperhey (Netherlands) applies for a patent on a "spyglass"
1609–10	**Galileo** builds his own ($\sim 20\times$): Jupiter's 4 large moons, the phases of Venus, lunar mountains, countless Milky Way stars
1611	**Kepler** designs the convex-eyepiece refractor still used today
1668	**Isaac Newton** builds the first practical **reflecting telescope**, free of colour fringing
1672	Laurent Cassegrain proposes the folded design with a convex secondary mirror
1781	William Herschel discovers Uranus with a home-made reflector
1990	**Hubble Space Telescope** launched (2.4 m mirror)
2021	**James Webb Space Telescope** launched (6.5 m segmented mirror)

2. What a telescope actually does

In order of importance:

1. **Gathers light.** A bigger opening (**aperture** D) collects more light, so you see **fainter** objects. *This is the most important specification.*
2. **Resolves detail.** A bigger aperture also shows **finer detail** — close double stars, Jupiter's cloud bands.
3. **Magnifies.** The eyepiece enlarges the image — the **least important** number.

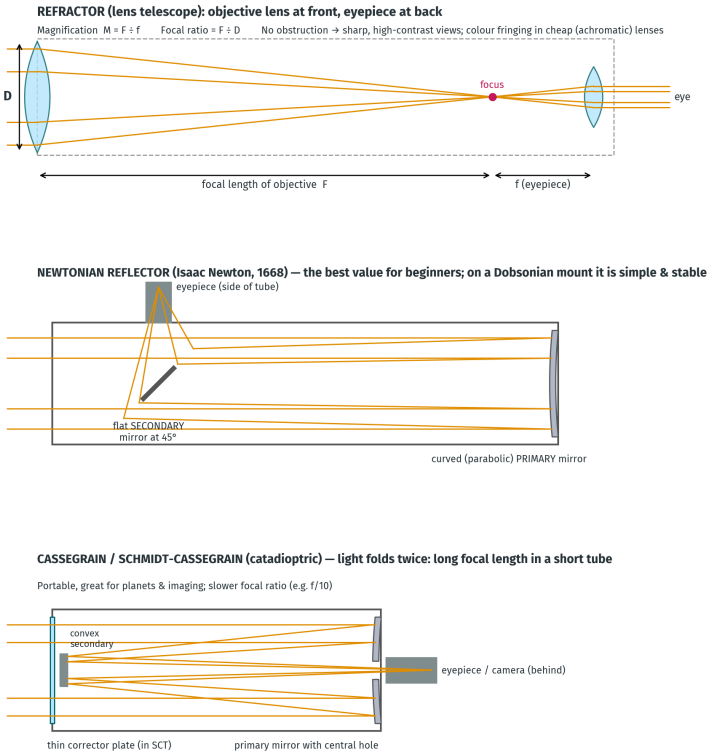
Beware of advertising

A box saying "**525× POWER!**" on a 60 mm telescope is misleading: the image becomes a dim blur long before that. As a rule of thumb, a telescope cannot usefully magnify more than about **twice its aperture in millimetres** (about 120× for 60 mm). **Always compare telescopes by aperture.**

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3. The three families of telescopes



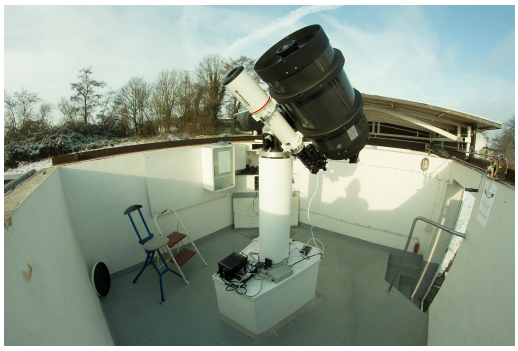
Light paths in a refractor, a Newtonian reflector and a Schmidt-Cassegrain telescope.



Refractor on a German equatorial mount (Gustaaf Prins, CC BY 2.0)



Dobsonian (Newtonian on a box mount) (James Stewart, public domain)



Schmidt-Cassegrain, 35 cm (WHS-2V, CC BY-SA 4.0)

Photos via Wikimedia Commons.

A. Refractors (lens telescopes)

A large **objective lens** at the front bends (refracts) light to a focus; an **eyepiece** magnifies the image. **Achromatic** refractors use two lens elements to reduce **chromatic aberration** — coloured fringes caused because glass bends blue and red light by different amounts; cheap ones still show purple halos. **Apochromatic (APO)** refractors use special glass and are almost colour-free, but expensive.

Pros	Cons
sharp, high-contrast images (no obstruction); sealed tube, little maintenance; quick to set up — great for the Moon, planets and double stars	expensive per mm of aperture; colour fringing (achromats); long tubes for large apertures

B. Reflectors (mirror telescopes)

A **concave primary mirror** at the bottom of the tube reflects light to a focus. In a **Newtonian**, a small **flat secondary mirror** at 45° sends the light out of the **side** of the tube. A **Dobsonian** is a Newtonian on a simple wooden box mount (John Dobson, 1960s) — a 150–200 mm "Dob" is the classic best first telescope for a club or serious beginner.

Pros	Cons
most aperture for your money; no chromatic aberration; easy to make — many amateurs grind their own mirrors	open tube collects dust; mirrors need occasional **collimation** (alignment); secondary mirror blocks a little light; needs time to cool down

C. Catadioptric (mirror + lens) telescopes

Schmidt-Cassegrain (SCT) and **Maksutov** designs have a front corrector lens, a primary mirror with a hole, and a convex secondary that reflects the light back through the hole — the light path is **folded twice**.

Pros	Cons
long focal length in a short, portable tube; excellent for planets, the Moon and photography; often sold on GoTo mounts	more expensive than a Newtonian of the same aperture; narrow field; the corrector dews up in humid coastal India — use a dew shield

Type	Typical beginner size	Best for	Cost
Achromatic refractor	70–102 mm	Moon, planets, travel	low–medium
Newtonian / Dobsonian	114–254 mm	everything, especially faint deep-sky objects	low–medium (best value)
Schmidt-Cassegrain	127–235 mm	planets, imaging, portability	high
Binoculars (7 × 50, 10 × 50)	50 mm	**learning the sky**, Milky Way, clusters	low

Advice

A good pair of 10×50 **binoculars** is often a better first instrument than a cheap telescope: "10×50" means 10× magnification and 50 mm lenses.

4. Parts of a telescope

Part	What it does
Objective (lens or primary mirror)	collects light and forms an image at the focus; its diameter is the **aperture** D, its focal length F
Focal length F	distance from objective to focus; long F → high magnification, narrow field
Focal ratio	$N = F/D$; e.g. f/5 is "fast" (wide fields, photography), f/10 "slow" (high power, planets)
Eyepiece	small magnifier with its own focal length f (25 mm, 10 mm…); **shorter f = more magnification**; also has an apparent field of view (50°–82°)
Barlow lens	placed before the eyepiece; multiplies magnification (usually 2×)

(continued)

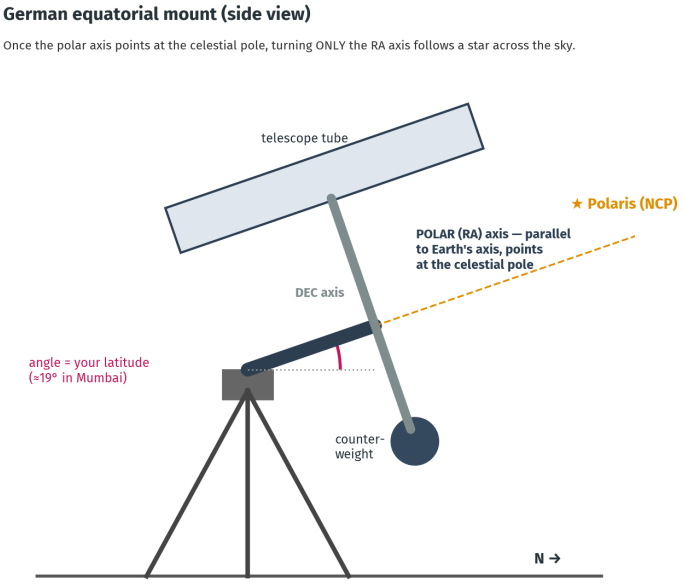
Part	What it does
Star diagonal	90° mirror/prism for comfortable viewing in refractors and SCTs (flips the image left–right)
Focuser	moves the eyepiece in and out to focus
Finder	low-power finderscope or red-dot finder for aiming — must be aligned with the main telescope
Mount & tripod	holds the telescope steady and lets it move — a wobbly mount ruins a good telescope
Filters	Moon filter; light-pollution filters; a certified **full-aperture solar filter** (front of the telescope only!)

Safety first

NEVER point a telescope or finder at the Sun without a **certified solar filter covering the FRONT** of the telescope, and cap the finder. Small "sun filters" that screw into eyepieces can crack from the heat and cause instant blindness — never use them.

5. Mounts

Mount	How it moves	Pros	Cons
Alt-azimuth	up–down and left–right	simple, intuitive	must adjust both axes to follow a star
Dobsonian	alt-az on a rocker box	very stable, cheap, quick	not for long-exposure photography
Equatorial (e.g. German)	one axis (polar/RA axis) points at the celestial pole	follow stars by turning one axis; motor tracking	heavier; needs polar alignment and counterweights
GoTo	computer + motors	finds thousands of objects, tracks	needs power and alignment; you may not learn the sky



A German equatorial mount: the polar axis is tilted at your latitude and points at the celestial pole.

6. India's telescopes for research — and for everyone



The 3.6 m Devasthal Optical Telescope, ARIES, Nainital (Ajay Talwar, CC BY 4.0)



One of the 30 dishes of the GMRT near Pune (DesiBoy101, CC BY 4.0)

Facility	Where	What
Kodaikanal Solar Observatory (1899)	Tamil Nadu	over a century of daily Sun images; the Evershed effect in sunspots discovered here (1909)
Vainu Bappu Telescope, 2.34 m (1986)	Kavalur, TN	once Asia's largest optical telescope
Himalayan Chandra Telescope, 2.0 m (2001)	Hanle, Ladakh	at 4,500 m, operated remotely from Bengaluru
Devasthal Optical Telescope, 3.6 m (2016)	near Nainital	**India's largest optical telescope**
Giant Metrewave Radio Telescope	Khodad, near Pune	30 dishes of 45 m — a world-leading low-frequency radio telescope
AstroSat (2015); Aditya-L1 (2023)	in space	India's multi-wavelength observatory; India's first solar mission

India also partners in the Thirty Meter Telescope (TMT) and the Square Kilometre Array (SKA). Many cities have planetaria and public observatories (the Birla planetaria, Nehru planetaria, science centres) that run telescope nights — visit one!

Key points to remember

- Telescope jobs: gather light > resolve detail > magnify. Compare telescopes by **aperture**.
- Refractor: lens, sharp, colour fringing. Newtonian/Dobsonian: mirror, best value. SCT/Maksutov: folded, compact, long focal length.
- Focal ratio $N = F/D$; shorter eyepiece focal length = more magnification.
- Mounts: alt-az / Dobsonian (simple), equatorial (polar axis at your latitude), GoTo (computerised).
- Never look at the Sun without a certified front solar filter. India's largest optical telescope: 3.6 m Devasthal.

Check yourself

1. Why is "675× magnification!" a bad reason to buy a telescope?
2. What causes chromatic aberration, and which telescope type has none?
3. What is the focal length of a 100 mm f/8 telescope?
4. Why must an equatorial mount's polar axis point at the celestial pole?

Answers: 1. Useful magnification is limited by aperture ($\sim 2 \times D$ in mm) and the atmosphere. 2. Glass bends colours differently; reflectors have none. 3. $F = 8 \times 100 = 800$ mm. 4. So that turning one axis exactly matches the sky's rotation about the pole.

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §6.1 "Telescopes" and §6.2 "Telescopes Today" — `https://openstax.org/books/astronomy-2e/pages/6-1-telescopes`
- NCERT *Physics* Class XII, Part II, Ch. 9 "Ray Optics and Optical Instruments" (the telescope section) — `https://ncert.nic.in/textbook.php`
- G. Consolmagno & D. M. Davis, *Turn Left at Orion* (Cambridge University Press) — the classic small-telescope guide.